

GPU-accelerated single-cell and spatial transcriptomics on NVIDIA DGX H100

A systematic benchmark of speed, scalability, and biological concordance

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The scale problem

- Single-cell RNA-seq datasets now routinely exceed **1 million cells**
- Spatial platforms (Visium HD) produce **hundreds of thousands of bins** per section
- The standard tools (**Scanpy**, **Squidpy**) are **CPU-bound**
- A full atlas-scale analysis can take **hours to days**

The question

Can GPUs make atlas-scale single-cell analysis routine — **without changing the biology?**

What we benchmarked

Hardware

- NVIDIA **DGX H100** node: 8× H100 80 GB, 112 cores, 2 TB RAM
- Spatial: local **RTX 4090** workstation

Software

- CPU: **Scanpy 1.12** (100 threads)
- GPU: **rapids-singlecell 0.14.1** (RAPIDS / CuPy / cuGraph)
- Reproducible **Singularity** containers

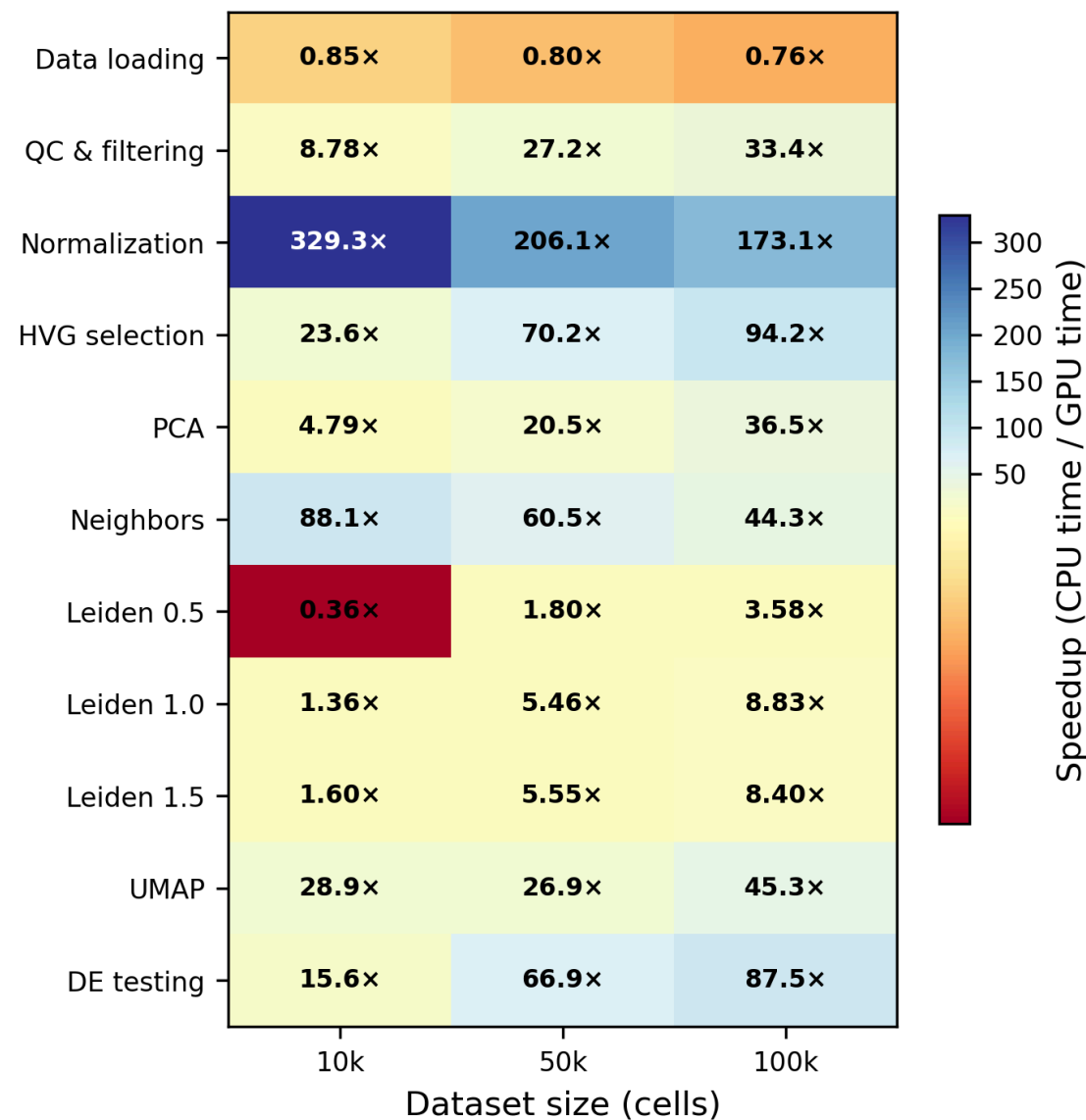
Five questions

1. Per-step & end-to-end **speedup**
2. **Multi-GPU** scaling (1→8 H100)
3. Biological **concordance** (CPU vs GPU)
4. **Maximum capacity** on one node
5. Generalisation to **spatial** transcriptomics

The pipeline is identical on both sides

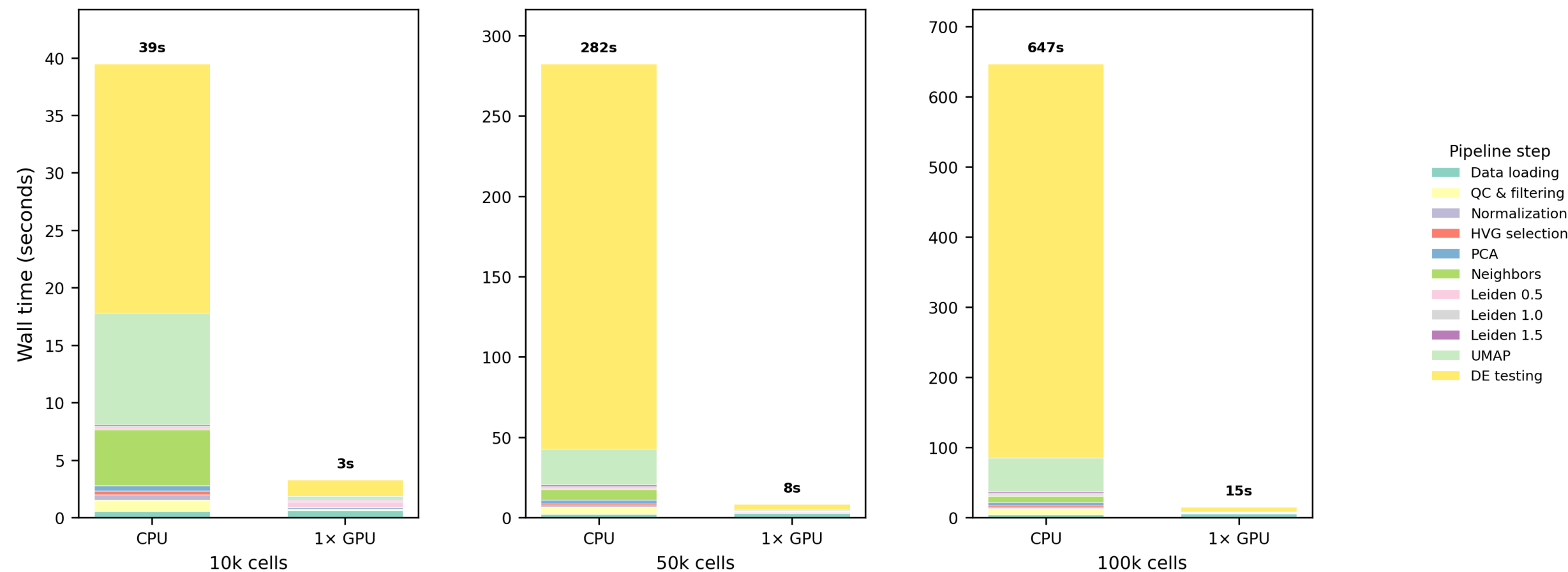
- Same dataset: **10x 1.3M mouse brain cells**, subsampled to 10k / 50k / 100k / 500k / 1.3M
- Same parameters: QC → normalise → **2,000 HVGs** → PCA (50) → kNN (k=15) → **Leiden** → UMAP → **Wilcoxon DE**
- Same seed (42), **five independent repeats** per configuration
- Concordance measured directly: HVG Jaccard, PCA correlation, ARI/NMI, DE log-fold-change

Up to 120× faster end-to-end



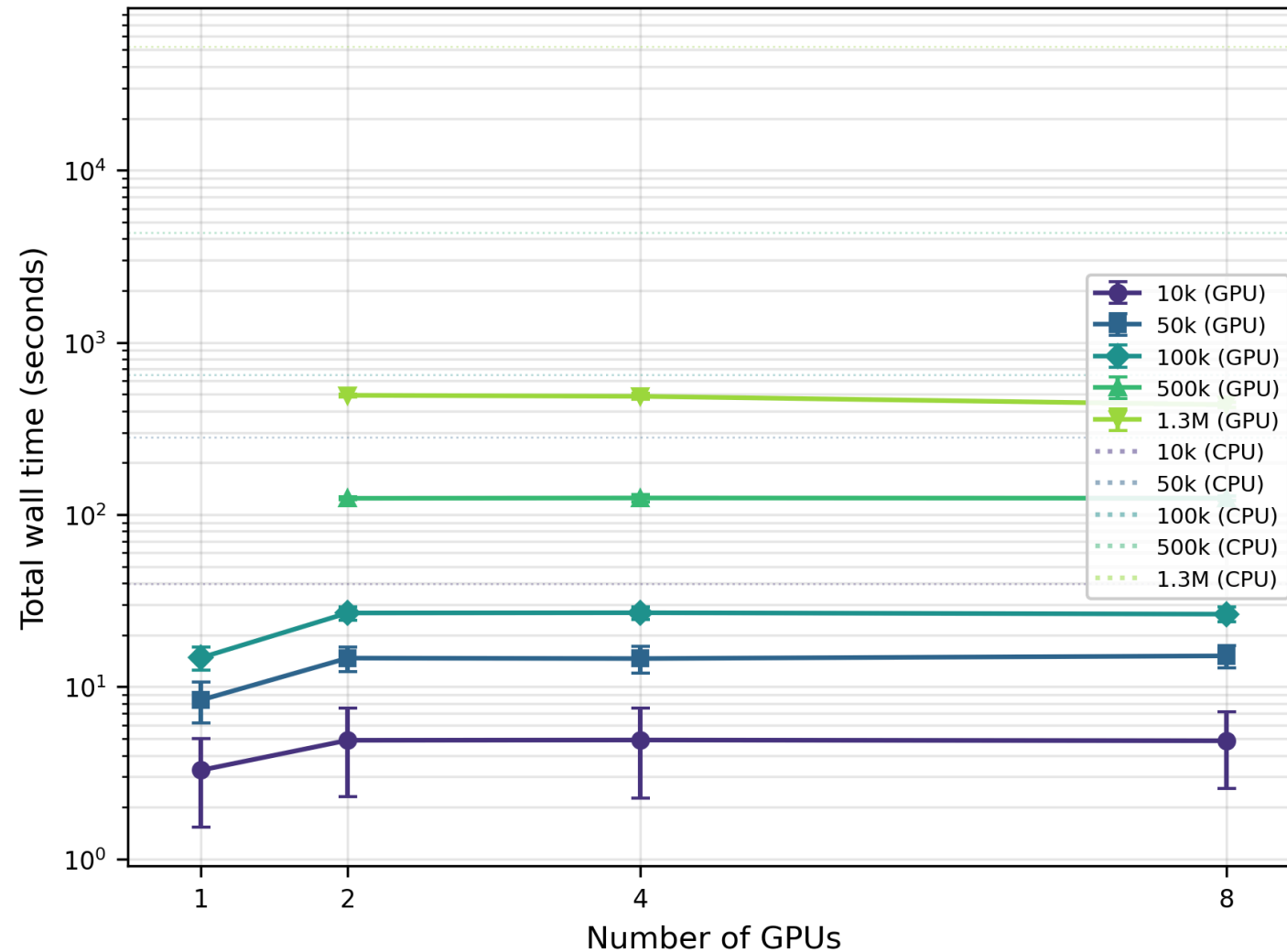
Per-step speedup of single-GPU rapids-singlecell vs CPU Scanpy. Normalisation reaches 329x; data loading is the one place CPU wins (fixed CUDA startup cost).

Where the time goes



Step-level wall-time on CPU vs single GPU. On CPU, differential expression and neighbours dominate; on GPU, every step collapses to seconds.

More GPUs help less than you'd hope



- 8 GPUs only **12% faster** than 2 GPUs at 1.3M cells
- In an 8-GPU run: **75%** of wall time is CPU preprocessing
- Only **4%** runs on multiple GPUs
- Classic **Amdahl's law**

Total time vs number of GPUs. Curves are nearly flat beyond 2 GPUs.

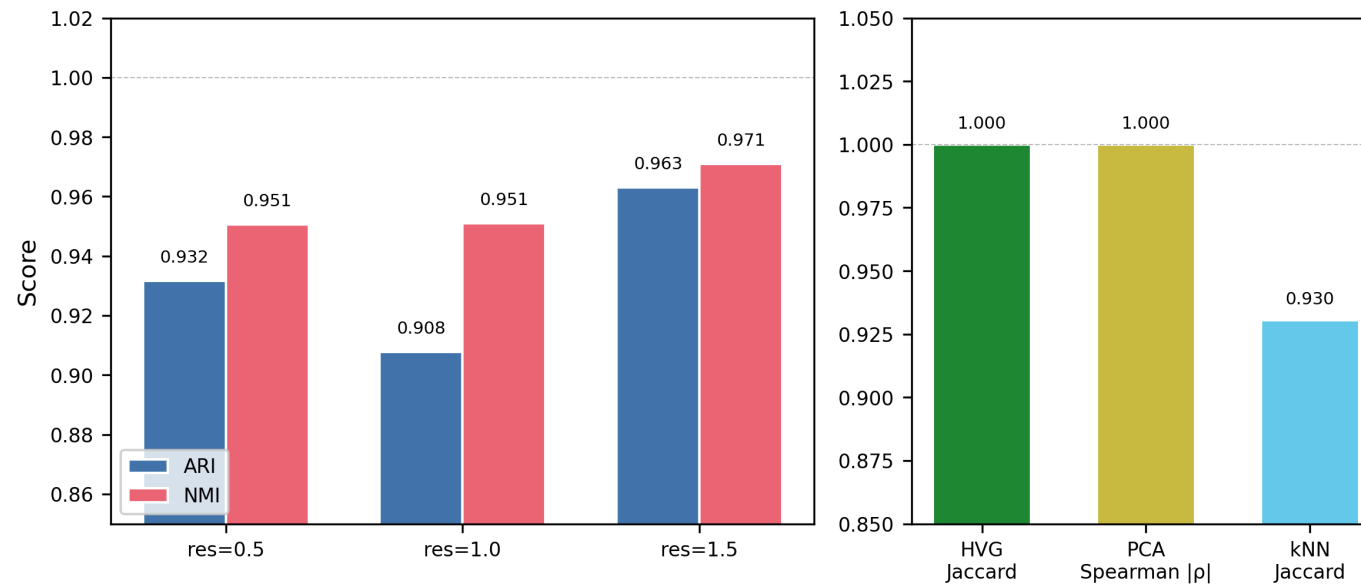
So use the node differently

Practical takeaway

A DGX node is better used as **eight independent single-GPU workers** running eight analyses in parallel — than as one 8-GPU worker on a single analysis.

- Most of the value is on **one** GPU
- A single cloud-rented H100 (~\$2–4/hour) captures nearly all the benefit
- GPU acceleration is **accessible**, not just for those with a DGX

The biology is preserved

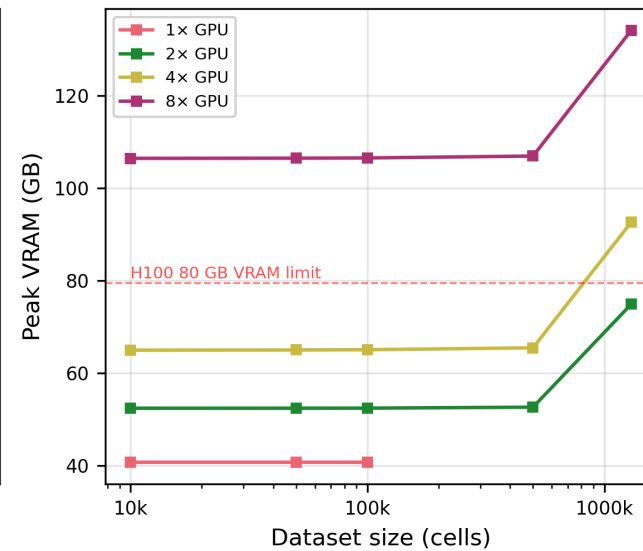
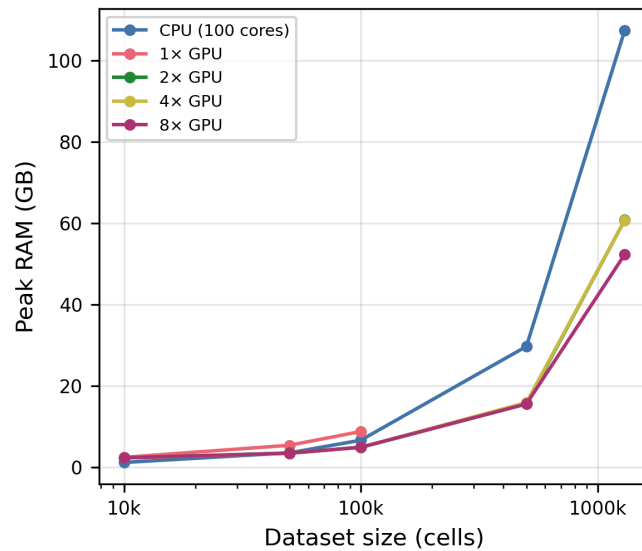


- HVG selection **identical** (Jaccard = 1.0)
- PCA loadings **perfectly correlated** ($|\rho| = 1.0$)
- Leiden **ARI 0.908–0.963**, NMI 0.951–0.971
- DE log-fold-change **$p = 0.946$**

Speed without changing conclusions.

CPU–GPU concordance on 10k cells: clustering ARI/NMI and feature-level agreement.

How far can one node go?



- Succeeded at **11.9M cells** on one DGX node
- Failed at 12.0M (Leiden OOM)
- The wall is **CPU RAM (535 GB)**, not GPU VRAM
- Aggregate VRAM only **7.6%** of the 640 GB available

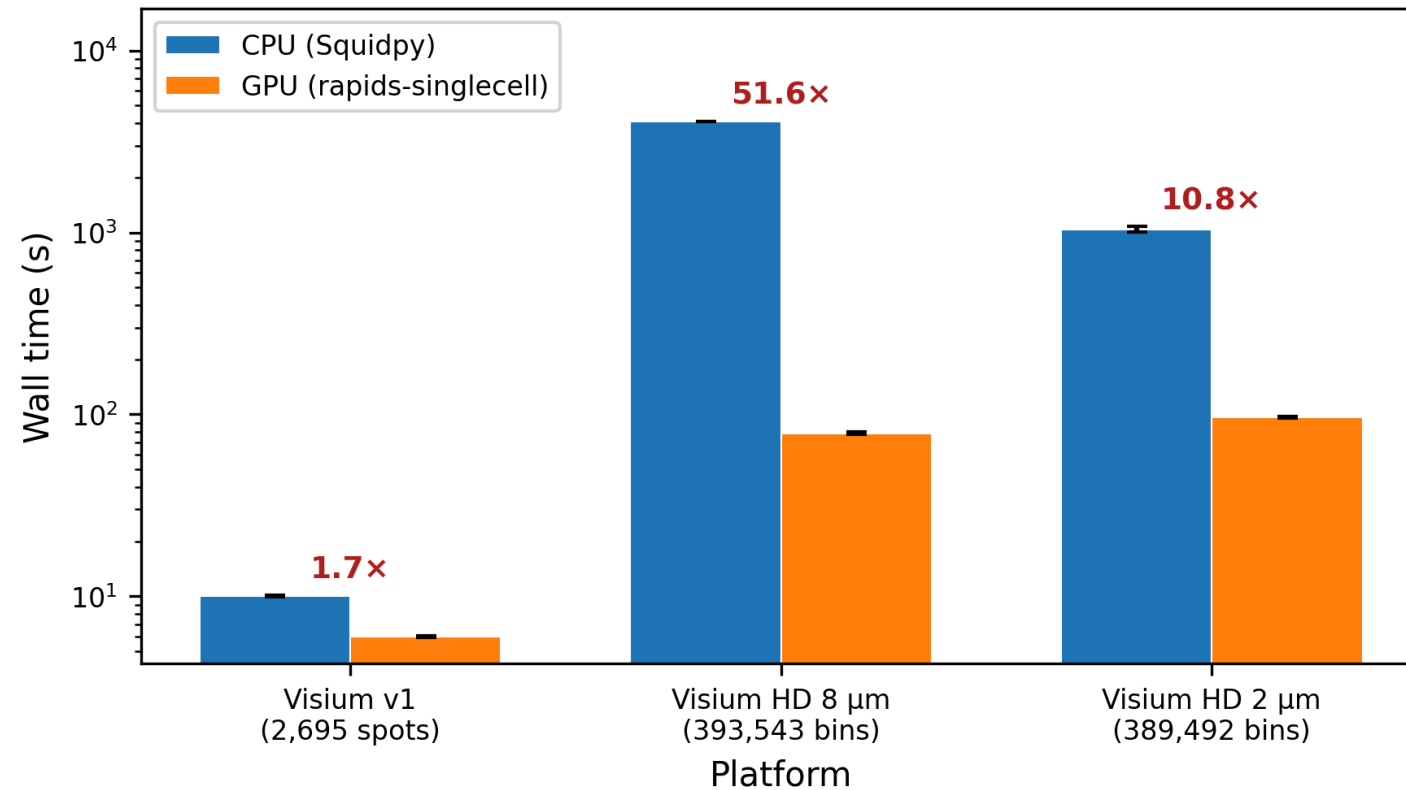
Memory profile: GPU VRAM stays flat and low; CPU RAM is the constraint.

Differential expression: pseudo-bulk wins twice

Method	Time @ 3.4M cells	Note
CPU t-test (cell-level)	5,598 s	baseline
GPU Wilcoxon	826 s	
Pseudo-bulk aggregation	128 s	44× faster

- Pseudo-bulk is the **fastest** *and* the most **statistically rigorous** (avoids pseudoreplication)
- Fast and correct are not in tension here

It generalises to spatial transcriptomics



- Up to **51.6x** end-to-end (Visium HD 8 μm)
- Co-occurrence analysis: **3,272x**
- Spatial autocorrelation concordance $\rho \geq 0.9995$
- On a **single consumer RTX 4090**

End-to-end and per-operation speedup across three Visium platforms.

Takeaways

1. **120× faster** end-to-end, with **biologically concordant** results
2. Bottleneck is **CPU preprocessing**, not GPU compute — so most value is on **one GPU**
3. Scaling limit is **CPU RAM**, not VRAM (11.9M cells on one node)
4. **Pseudo-bulk DE** is fastest *and* most rigorous
5. Generalises to **spatial** (51.6×) — even on consumer hardware

Note

Code, containers, and all benchmark data: github.com/lucavd/2026.scRNA_DGX

Thank you

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